

Tanay Pratap Singh

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Master's of Science in Applied Data Science	Syracuse University, Syracuse, NY	Jan. 2025 – Dec. 2026
B.Tech in Computer Science and Engineering	JSSATE Noida, UP, India	Dec. 2020 – May 2024

EXPERIENCE

Research Assistant

Feb. 2026 - Present

Syracuse University

Syracuse, NY

- Parsed 276,000+ Zooniverse volunteer classifications for the project Gravity Spy 2.0 (NSF-funded) using Python and pandas, reconciling 5 undocumented metadata schemas across 25,104 LIGO glitch subjects and 8,293 auxiliary channels.
- Reverse-engineered the Zooniverse Panoptes API to build a batch pipeline retrieving 49,000 spectrograms; processed 1200x1200 mosaics into paired 224x224 channel heatmaps with OpenCV.
- Trained a two-input CNN with a shared MobileNetV2 backbone in TensorFlow/Keras to predict volunteer similarity labels from paired strain and auxiliary-channel spectrograms.
- Built a glitch-type by auxiliary-channel similarity matrix from model outputs to identify detector subsystems (PEM, SUS, LSC, ISI, ASC, CAL) causally linked to specific glitch types.

LSS Consultant

Jan. 2026 - Present

JMA Wireless

Liverpool, NY

- Designed an executive Power BI dashboard comparing JMA's two Jumper assembly lines (Jumper 1 manual, Jumper 2 fully automated), consolidating 4 operational KPIs into a single decision support view for the SVP of Global Operations.
- Built a Python ETL pipeline to flatten nested JSON PLC records across 500,800+ rows, cutting Power BI memory load from 500+ GB to under 5 MB and enabling one click refresh for non technical users.
- Authored 10+ DMAIC artifacts (Charter, SIPOC, CTQC, FMEA, RASIC, Control Plan) and applied Minitab to baseline process performance.
- Partnered with a four person cross functional team and presented findings to JMA executive sponsors, translating analysis into business recommendations tied to the financial plan.

Cybersecurity Teaching Assistant

Summer 2025

Syracuse University

Syracuse, NY

- Developed and delivered 15+ hands-on cybersecurity labs covering on cryptography, network security, penetration testing, and digital forensics using Wireshark, Kali Linux, Nmap, OpenSSL, and Metasploit, which increased student confidence in applying security tools.
- Built and maintained multi-VM lab environments (Kali Linux, Windows 10/Server, CentOS, Metasploitable) to enable realistic attack simulations and vulnerability assessments, allowing students to practice exploitation techniques in a safe setting.
- Graded and reviewed security labs and incident analysis reports, reinforcing packet analysis, OSINT reconnaissance, ethical hacking and hashing (SHA-256, SHA-512) concepts, which helped students achieve clearer understanding of compliance requirements.

Data and Product Analyst Intern

Jul. 2023 – Aug. 2023

ProProfs

Noida, UP, India

- Performed customer and product analytics on 2,000+ client datasets using Python, SQL, and Excel to uncover behavior patterns and conversion drivers, enabling the product team to prioritize two high-impact features that lifted trial-to-paid conversion.
- Built Tableau dashboards visualizing feature usage and conversion trends across SaaS products serving 5,000+ users, surfacing under-used features that reduced churn by 5 percent and helped lift product ratings from 4.2 to 4.6.

TECHNICAL SKILLS

Languages and Frameworks: Python, SQL, R, JavaScript, HTML/CSS

ML and AI: Scikit-learn, TensorFlow/Keras, XGBoost, CatBoost, OpenCV, SHAP; Random Forest, Gradient Boosting, LSTM, CNN, NLP, Computer Vision, Feature Engineering, Bayesian Optimization, Model Deployment

Data and Analytics: Pandas, NumPy, Spark, Tableau, Power BI, Excel, Streamlit; EDA, A/B Testing, Time Series Forecasting, Hypothesis Testing, Statistical Analysis

Cloud and Data Engineering: Azure, MySQL, ETL Pipelines, JSON Data Engineering, Git/GitHub, Jupyter

Statistics, Modeling and Business: Linear/Logistic Regression, Multiple Regression, ANOVA, Hypothesis Testing, Time Series Forecasting, Multinomial/Ordered Logit, Conjoint Analysis, K-Means Clustering, Lean Six Sigma (DMAIC), SIPOC, Process Mapping, A/B Testing, Technical Documentation, Stakeholder Communication